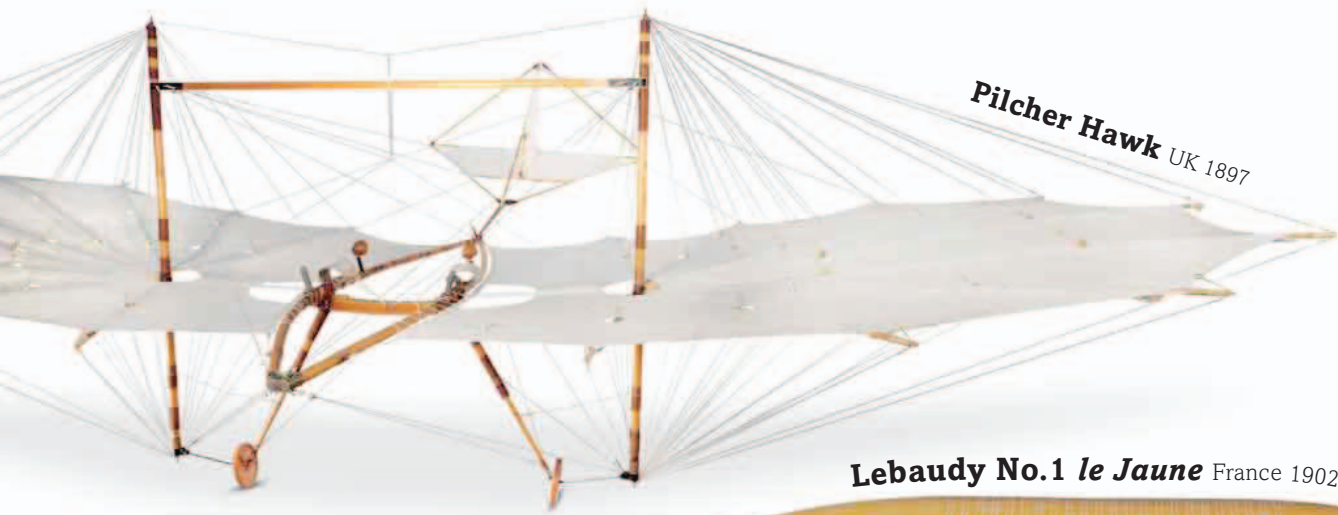
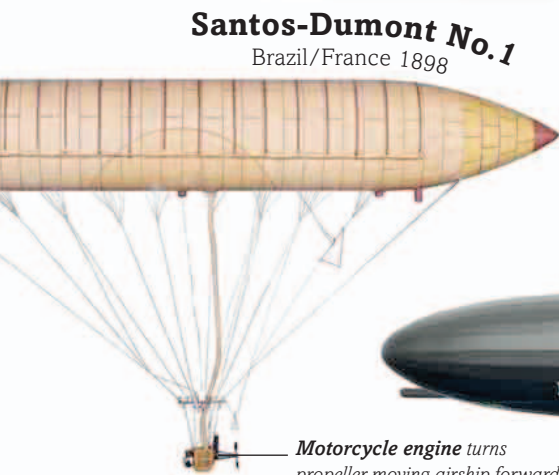
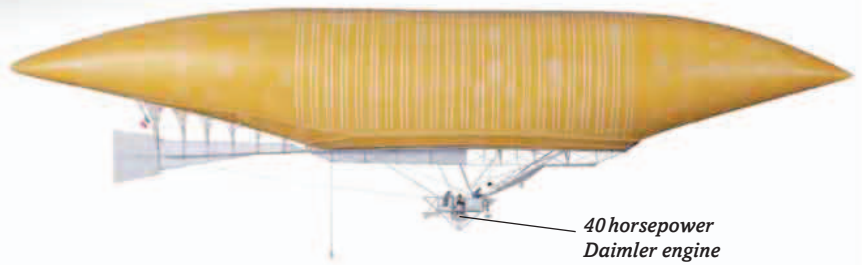




Pilcher Hawk UK 1897



Santos-Dumont No. 1
Brazil/France 1898



Lebaudy No. 1 le Jaune France 1902

40 horsepower
Daimler engine
turns two propellers



Zeppelin LZ 96 Germany 1917

Hinged rudder
helps steer the craft



LZ 127 Graf Zeppelin Germany 1928

Gondola holds crew
and up to 20 passengers

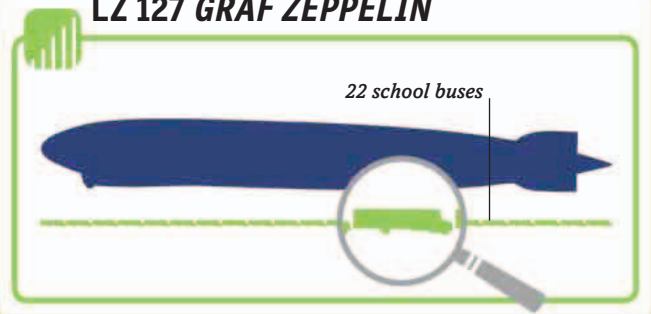
778-ft- (237-m-) long hull
covered in cotton over a rigid
frame of aluminum alloy girders



17-ft-4-in-
(5.3-m-)
long glider

Passengers
on the *Graf
Zeppelin* had their
own beds, a
main dining room,
and meals made
onboard.

LZ 127 GRAF ZEPPELIN



22 school buses

la Charlière, and, in 1898, the first flight of the airship **Santos-Dumont No. 1**. In Germany, large airships, such as the **Zeppelin LZ 96**, scouted and bombed during World War I, while postwar airships, such as the **Graf Zeppelin**, offered long-distance transportation to the

wealthy. Other inventors believed that winged gliders were the way up. In the 1890s, German engineer Otto Lilienthal made many successful flights in gliders such as the **Normal Apparatus**. His work inspired other glider designs, as well as the Wright Brothers' work on a powered aircraft.